



EE1005 – Digital Logic Design

Assignment Number 4

Fall 2022

Maximum Marks: 50

Due Date: 29 November, 2022

Submitted By

Name: _____

Student ID: _____

Section: _____

Submitted To

Dr. Adil Zulfiqar
Assistant Professor & In-charge– EE Department

Submission Date

____ November 2022

Question Number	1	2	3	4	5	Total
Total Marks	15	10	10	10	05	50
Obtained Marks						

1. A sequential circuit with two D flip-flops A and B , two inputs, x and y ; and one output z is specified by the following next-state and output equations

$$A(t + 1) = xy' + xB$$

$$B(t + 1) = xA + xB'$$

$$z = A$$

- (a) Draw the logic diagram of the circuit.
 - (b) List the state table for the sequential circuit.
 - (c) Draw the corresponding state diagram.
2. A sequential circuit has one flip-flop Q , two inputs x and y , and one output S . It consists of a full-adder circuit connected to a D flip-flop, as shown in Fig. Derive the state table and state diagram of the sequential circuit.

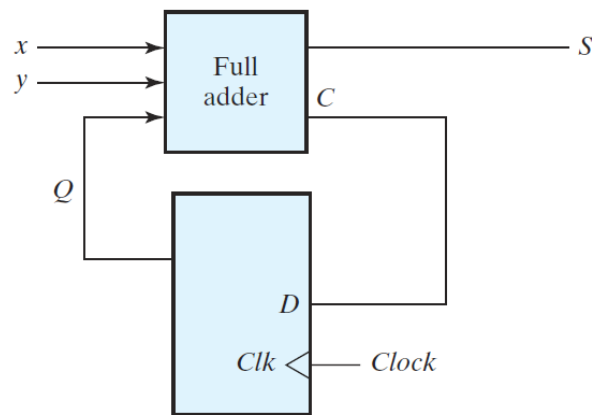
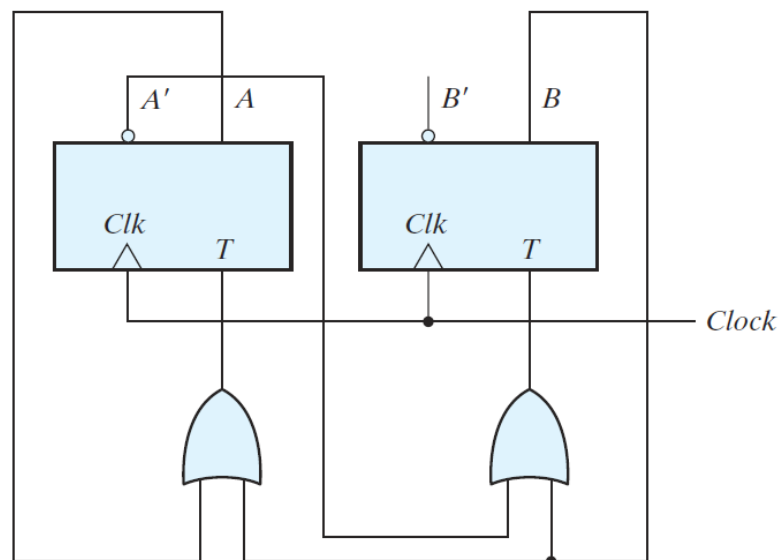


Fig.

3. Derive the state table and the state diagram of the sequential circuit shown in Fig. Explain the function that the circuit performs.



Fig

4. A sequential circuit has two JK flip-flops A and B and one input x . The circuit is described by the following flip-flop input equations:

$$J_A = x \quad K_A = B$$

$$J_B = x \quad K_B = A'$$

- (a) Derive the state equations $A(t+1)$ and $B(t+1)$ by substituting the input equations for the J and K variables.
 (b) Draw the state diagram of the circuit.

5. For the following state table

Present State	Next State		Output	
	$x = 0$	$x = 1$	$x = 0$	$x = 1$
a	f	b	0	0
b	d	c	0	0
c	f	e	0	0
d	g	a	1	0
e	d	c	0	0
f	f	b	1	1
g	g	h	0	1
h	g	a	1	0

Draw the corresponding state diagram.